

Ex# 4.3

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(25)

Q Find the A.M b/w:

(i) 18 & 26

Sol

let A is the A.M
b/w 18 & 26.

$$\therefore A = \frac{a+b}{2}$$

Here, $a = 18$ & $b = 26$

$$\therefore A = \frac{18+26}{2}$$

$\Rightarrow A = 22$ is the re-
quired A.M b/w 18 & 26.

(ii) $10+5\sqrt{3}$ and $4-5\sqrt{3}$

Sol

let A be the A.M b/w
 $10+5\sqrt{3}$ and $4-5\sqrt{3}$

$$\therefore A = \frac{a+b}{2}$$

Here, $a = 10+5\sqrt{3}$ & $b = 4-5\sqrt{3}$

$$\therefore A = \frac{10+5\sqrt{3}+4-5\sqrt{3}}{2}$$

$$\Rightarrow A = \frac{14}{2} = 7$$

It is the required A.M b/w

$$10+5\sqrt{3} \text{ and } 4-5\sqrt{3}$$

Sol

(iii) $3a+2b$ and $5a-6b$

Sol

let A be the A.M b/w
 $3a+2b$ and $5a-6b$

$$\therefore A = \frac{a+b}{2}$$

Here, $a = 3a+2b$ & $b = 5a-6b$

$$\therefore A = \frac{3a+2b+5a-6b}{2}$$

$$\Rightarrow A = \frac{8a-4b}{2} = \frac{2(4a-2b)}{2}$$

$\Rightarrow A = 4a-2b$ is the
required A.M.

(iv) $5+7\sqrt{5}$ and $8-7\sqrt{5}$

Sol

let A be the A.M b/w
 $5+7\sqrt{5}$ and $8-7\sqrt{5}$

$$\therefore A = \frac{a+b}{2}$$

Here, $a = 5+7\sqrt{5}$ and $b = 8-7\sqrt{5}$

$$\therefore A = \frac{5+7\sqrt{5}+8-7\sqrt{5}}{2}$$

$\Rightarrow A = \frac{13}{2}$ is the required
A.M.

(vi) $3a-5$ and $5a+3$
Sol

let A be the A.M
 b/w $3a-5$ & $5a+3$

$$\therefore A = \frac{a+b}{2}$$

Here, $a = 3a-5$ & $b = 5a+3$

$$\therefore A = \frac{3a-5+5a+3}{2}$$

$$\Rightarrow A = \frac{8a-2}{2} = 2(4a-1)$$

$\Rightarrow A = 4a-1$ is the
 required A.M. Ans

(26)
 $3a^2-5a+6$ & $-a^2+7a-4$
Sol

let A be the A.M b/w
 $3a^2-5a+6$ and $-a^2+7a-4$

$$\therefore A = \frac{a+b}{2}$$

here,

$a = 3a^2-5a+6$ & $b = -a^2+7a-4$

$$\therefore A = \frac{3a^2-5a+6-a^2+7a-4}{2}$$

$$\Rightarrow A = \frac{2a^2+2a+2}{2} = 2(a^2+a+1)$$

$\Rightarrow A = a^2+a+1$ is the
 required A.M. Ans

(3) Insert three A.M b/w 3 and 11

Sol

let A_1, A_2 and A_3 are three A.M b/w 3 and 11
 i.e., $3, A_1, A_2, A_3, 11$

$$\therefore d = \frac{b-a}{n+1}$$

here, $a = 3, n = 3$ & $b = 11$

$$\therefore d = \frac{11-3}{3+1}$$

$$\Rightarrow d = 8/4$$

$$\Rightarrow d = 2$$

Now,

$$\Rightarrow A_1 = a + d = 3 + 2 = 5$$

$$\Rightarrow A_2 = a + 2d = 3 + 2(2) = 7$$

$$\Rightarrow A_3 = a + 3d = 3 + 3(2) = 9$$

Thus 5, 7 and 9 are 3
 required A.M b/w
 3 and 11 Ans

③ Insert four A.Ms b/w $\sqrt{2}$ and $12/\sqrt{2}$
Sol

let A_1, A_2, A_3 , and A_4 are the 4 A.Ms.

$$\therefore d = \frac{b-a}{n+1}$$

here, $a = \sqrt{2}$, $b = 12/\sqrt{2}$ and $n = 4$

$$\therefore d = \frac{\frac{12}{\sqrt{2}} - \sqrt{2}}{4+1} = \frac{5\sqrt{2}}{5} = \sqrt{2}$$

Now,

$$A_1 = a + d = \sqrt{2} + \sqrt{2} = 2\sqrt{2}$$

$$A_2 = a + 2d = \sqrt{2} + 2(\sqrt{2}) = 3\sqrt{2}$$

$$A_3 = a + 3d = \sqrt{2} + 3\sqrt{2} = 4\sqrt{2}$$

$$A_4 = a + 4d = \sqrt{2} + 4\sqrt{2} = 5\sqrt{2}$$

④ Insert 5 A.Ms b/w $\sqrt{5}/2$ and $4\sqrt{5}$
Sol

let A_1, A_2, A_3, A_4 and A_5 are 5 A.Ms.

$$\therefore d = \frac{b-a}{n+1}$$

here, $a = \sqrt{5}/2$, $b = 4\sqrt{5}$ and $n = 5$

$$\therefore d = \frac{4\sqrt{5} - \sqrt{5}/2}{5+1} = \frac{7\sqrt{5}/2}{6} = \frac{7\sqrt{5}}{12}$$

Now,

$$A_1 = a + d = \sqrt{5}/2 + 7\sqrt{5}/12 = 13\sqrt{5}/12$$

$$A_2 = a + 2d = \sqrt{5}/2 + 2(7\sqrt{5}/12) = 5\sqrt{5}/3$$

(28)

$$A_3 = a + 3d = \sqrt{5}/2 + 3(7\sqrt{5}/12) = 9\sqrt{5}/4$$

$$A_4 = a + 4d = \sqrt{5}/2 + 4(7\sqrt{5}/12) = 17\sqrt{5}/6$$

$$A_5 = a + 5d = \sqrt{5}/2 + 5(7\sqrt{5}/12) = 41\sqrt{5}/12$$

Q.

⑤ Insert 6 A.Ms b/w -30 and 30

Sol

let A_1, A_2, A_3, A_4, A_5 and A_6 are the 6 A.Ms.

$$\therefore d = \frac{b-a}{n+1}$$

here, $a = -30$, $b = 30$ and $n = 6$

$$\therefore d = \frac{30 - (-30)}{6+1} = \frac{60}{7}$$

Now,

$$A_1 = a + d = -30 + \frac{60}{7} = -210 + 60/7 = -150/7$$

$$A_2 = a + 2d = -30 + 2(\frac{60}{7}) = -210 + 120/7 = -90/7$$

$$A_3 = a + 3d = -30 + 3(\frac{60}{7}) = -210 + 180/7 = -30/7$$

$$A_4 = a + 4d = -30 + 4(\frac{60}{7}) = -210 + 240/7 = 30/7$$

$$A_5 = a + 5d = -30 + 5(\frac{60}{7}) = -210 + 300/7 = 90/7$$

$$A_6 = a + 6d = -30 + 6(\frac{60}{7}) = -210 + 360/7 = 150/7$$

Q.

⑥ If 5 and 8 are two A.Ms b/w c and d , find c and d

Sol

Given: $c, 5, 8, d$

We can write it as: c, A_1, A_2, d

$$\therefore A_1 = 5 \text{ and } A_2 = 8$$

$$A_1 = \frac{c + A_2}{2} \Rightarrow \frac{5 = c + 8}{2} \Rightarrow 10 = c + 8 \Rightarrow c = 2$$

$$A_2 = \frac{A_1 + d}{2} \Rightarrow \frac{8 = 5 + d}{2} \Rightarrow 16 = 5 + d \Rightarrow d = 11$$

⑦ Find n so that $\frac{a^{n+5} + b^{n+5}}{a^{n+4} + b^{n+4}}$ may be A.M b/w a and b .

Sol

Since it is given that:

$$\Rightarrow \text{A.M} = \frac{a^{n+5} + b^{n+5}}{a^{n+4} + b^{n+4}} \quad \text{--- (1)}$$

As we know that,

$$\Rightarrow \text{A.M} = \frac{a+b}{2} \quad \text{--- (2)}$$

By comparing (1) and (2) we get,

$$\Rightarrow \frac{a+b}{2} = \frac{a^{n+5} + b^{n+5}}{a^{n+4} + b^{n+4}}$$

$$\Rightarrow \frac{a+b}{1+1} = \frac{a^{n+5} + b^{n+5}}{a^{n+4} + b^{n+4}}$$

$$\Rightarrow \frac{a+b}{a^0+b^0} = \frac{a^{n+5} + b^{n+5}}{a^{n+4} + b^{n+4}}$$

Since the bases are same therefore there powers should equal to each other-

$$\Rightarrow \frac{1+1}{0+0} = \frac{n+5+n+5}{n+4+n+4}$$

$$\Rightarrow \frac{2}{6} = \frac{2n+10}{2n+8}$$

So,

$$\Rightarrow 2n+10=2$$

$$\Rightarrow 2n = -8$$

$$\Rightarrow n = -4$$

Ex # 4.4

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① Sum the series.

① $3+8+13+\dots$ to 16 terms.

Sol

$$\therefore S_n = \frac{n}{2} \{2a + (n-1)d\}$$

here, $a=3$, $n=16$

$$d = 8-3 = 13-8 = 5$$

$$\therefore S_{16} = \frac{16}{2} \{2(3) + (16-1)5\}$$

$$\Rightarrow S_{16} = 8(6 + 75)$$

$$\Rightarrow S_{16} = 648$$

(ii) $(-3)+(-1)+1+3+5+\dots$ to 18 term

Sol

$$\therefore S_n = \frac{n}{2} \{2a + (n-1)d\}$$

here, $a=-3$, $n=18$

$$d = -1 - (-3) = -1 - (-1) = 2$$

$$\therefore S_{18} = \frac{18}{2} \{2(-3) + (18-1)2\}$$

$$\Rightarrow S_{18} = 9(-6 + 34)$$

$$\Rightarrow S_{18} = 252$$

(iii) $4.57+4.87+5.17+\dots$ to 22 term

Sol

$$\therefore S_n = \frac{n}{2} \{2a + (n-1)d\}$$

here, $a=4.57$, $n=22$ and

$$d = 4.87 - 4.57 = 5.17 - 4.87 = 0.3$$

$$\therefore S_{22} = \frac{22}{2} \{2(4.57) + (22-1)0.3\}$$

$$\Rightarrow S_{22} = 11(9.14 + 6.3)$$

$$\Rightarrow S_{22} = 11(15.44)$$

$$\Rightarrow S_{22} = 169.84$$